

because $f(P, S \cap M) \subset U$. Since S_p is subspace-representable under P , we have

$$f(P, S_p \cap C_{stb}) = f(P, S \cap M) \cap f(P, C_{stb}) = U \cap f(P, C_{stb}). \quad (8)$$

Recall that

$$U = \left\{ X \in \mathbb{R}(s)^{n_z \times n_w} \mid A^j \cdot X = 0 \quad \forall j = 1, \dots, n_z n_w - n \right\}$$

where A^j are as defined in Definition 11. Then U is a variety because it is defined by a system of affine polynomial equations in variables $x_{11}, \dots, x_{n_z n_w}$. Note that $\pi_{m+1}(\mathbf{V}(J')) \subset U$ by definition of $\mathbf{V}(J')$. Therefore, by Lemma 10, we have $\pi_{m+1}(\mathbf{V}(J)) \subset U$. Since $\mathbf{V}(J_{m+1})$ is the smallest variety containing $\pi_{m+1}(\mathbf{V}(J))$, we have $\mathbf{V}(J_{m+1}) \subset U$. Therefore

$$\pi_{m+1}(\mathbf{V}(J')) \subset \mathbf{V}(J_{m+1}) \subset U. \quad (9)$$

Now, note that $\pi_{m+1}(\mathbf{V}(J')) \cap f(P, C_{stb}) = U \cap f(P, C_{stb})$ by Lemma 9 and (8). Therefore, by (8) and (9)

$$f(P, S_p \cap C_{stb}) = \mathbf{V}(J_{m+1}) \cap f(P, C_{stb}).$$

We also have that $\mathbf{V}(J_{m+1}) \cap f(P, C_{stb})$ is convex since $U \cap f(P, C_{stb})$ is convex by convexity of U .

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Strict Lyapunov Functions for the Super-Twisting Algorithm

Jaime A. Moreno, *Member, IEEE*, and Marisol Osorio

Abstract—A method to construct a family of strict Lyapunov functions, i.e., with negative definite derivative, for the super-twisting algorithm, without or with perturbations, is provided. This second order sliding modes algorithm is widely used to design controllers, observers and exact differentiators. The proposed Lyapunov functions ascertain finite time convergence, provide an estimate of the convergence time, and ensure the robustness of the finite-time or ultimate boundedness for a class of perturbations wider than the classical ones for this algorithm. Since the Lyapunov functions and their derivatives are quadratic forms, the operation with them is as simple as for linear time invariant systems.

Index Terms—Discontinuous systems, Lyapunov functions, robust observers, second order sliding modes (SOSM).

I. INTRODUCTION

The *Super-Twisting Algorithm* (STA) is a well-known second order sliding modes (SOSM) algorithm introduced in [9], and it has been widely used for control [1], [4], [7], [9], [11], [12], observation [5], and robust exact differentiation [10], [15]. The STA can be written as

$$\begin{aligned} \dot{x}_1 &= -k_1 |x_1|^{1/2} \text{sign}(x_1) + x_2 + \varrho_1(x, t) \\ \dot{x}_2 &= -k_2 \text{sign}(x_1) + \varrho_2(x, t) \end{aligned} \quad (1)$$

where x_i are the scalar state variables, k_i are gains to be designed, and ϱ_i are the perturbation terms. Under some conditions on $k_1, k_2(1)$ is robust against a bounded perturbation $\varrho_1(x, t) = 0, |\varrho_2(x, t)| \leq L$. Since the righthand side of (1) is discontinuous, its solutions will be understood in the sense of Filippov [6].

Finite time convergence and robustness for the STA has been proved by geometrical methods [12], or by means of the Homogeneity properties of the algorithm [11], [14]. $V_w(x) = k_2 |x_1| + (1/2)x_2^2$ is a weak Lyapunov function for the system (1) without perturbations, since $\dot{V}_w(x) = -k_1 k_2 |x_1|^{1/2}$ is only negative semidefinite. (Finite time) convergence can only be asserted by using a generalization of LaSalle's invariance principle for discontinuous systems [14], but it is not possible to provide robustness results, or to estimate the convergence time from it. By a detailed analysis of the weak Lyapunov function $\sqrt{V_w(x)}$ in [18] finite time and robust convergence for the STA is proved. A strict (or strong) Lyapunov function, i.e., a positive definite (p.d.) function with negative definite (n.d.) derivative, has been proposed in [16], from which an estimation of the convergence time for the perturbed STA is drawn. However, the form of the function makes it difficult to operate with it for applications or further developments.

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J. A. Moreno is with Coordinación de Eléctrica y Computación, Instituto de Ingeniería, Universidad Nacional Autónoma de México, 04510 México D.F., Mexico (e-mail: jmorenop@ii.unam.mx).

M. Osorio is with Escuela de Ingeniería, Grupo de Investigación en Automática y Diseño, Universidad Pontificia Bolivariana, Cir. 1 Num. 70-01, Medellín, Colombia (e-mail: mosorio@upb.edu.co).

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The objective of this technical note (from which [13] is a preliminary conference version) is to propose a method to construct a *class* of strict Lyapunov functions for the STA, that has the following distinguishing features compared to previous results: i) It provides a method to generate a whole *family* of strict Lyapunov functions. ii) The family of Lyapunov functions is very simple to work with, since they are *quadratic forms*, and with time derivatives that are also quadratic forms, whose matrices, in the unperturbed case, are related by an Algebraic Lyapunov Equation (ALE), in exactly the same manner as in the Linear Time Invariant case. iii) For each member of the family *finite time convergence* of the STA can be derived, and an estimation of the Convergence Time is a simple function of the Lyapunov function. iv) The class of perturbations, for which *finite time stability* can be established, is much larger than the classical ones [16], since here $\varrho_1(x, t) \neq 0$ in (1) is allowed. Moreover, for a larger class of perturbations ultimate boundedness ("practical" stability) can be ascertain. v) For the perturbed case a method to design the gains and to construct robust (quadratic) Lyapunov functions is also provided. vi) The method is also able to provide Chetaev's functions to prove instability.

The rest of the technical note is organized as follows. In Section II a method to construct quadratic Lyapunov functions for the unperturbed STA is presented, and finite time stability and instability of the STA can be proved by means of the Lyapunov functions. In Section III Lyapunov functions for the perturbed STA are constructed, from which a gain design method is derived. For different classes of perturbations finite-time stability and ultimate boundedness results are obtained. An illustrative Example is presented in Section IV. The Appendix contains all the proofs.

II. LYAPUNOV FUNCTIONS FOR THE STA WITHOUT PERTURBATIONS

Since the derivative of the vector $\zeta^T = [\zeta_1, \zeta_2] = [|x_1|^{1/2} \text{sign}(x_1), x_2]$ is given by

$$\dot{\zeta} = \frac{1}{|\zeta_1|} A \zeta, \quad A = \begin{bmatrix} -\frac{1}{2}k_1 & \frac{1}{2} \\ -k_2 & 0 \end{bmatrix} \quad (2)$$

where $|\zeta_1| = |x_1|^{1/2}$, it is appealing to consider the quadratic function

$$V(x) = \zeta^T P \zeta \quad (3)$$

with P a constant, symmetric and positive definite matrix, as a *strict* Lyapunov function candidate for the STA (1) without perturbations ($\varrho_1 = \varrho_2 = 0$). Compared to the Lyapunov function proposed in [16], $V(x)$ in (3) seems more simple and appropriate for applications, since it is a quadratic form in ζ , and its derivative along the trajectories of the unperturbed system (1) is

$$\dot{V} = -|x_1|^{-1/2} \zeta^T Q \zeta \quad (4)$$

(almost everywhere), where P and Q are related by the Algebraic Lyapunov Equation (ALE)

$$A^T P + P A = -Q. \quad (5)$$

Since A is Hurwitz iff $k_1 > 0, k_2 > 0$, for every $Q = Q^T > 0$ there exists a (unique) solution $P = P^T > 0$ of the ALE (5), so that $V(x)$ is a strict Lyapunov function. So our results resembles the classical results for Linear Time Invariant systems. Next Theorem 1 shows in fact, that the stability of the equilibrium $x = 0$ of (1) is completely determined by the stability of the matrix A : it is stable or unstable if and only if A is stable or unstable. This is a remarkable fact.

Although the results of Theorem 1 are very simple, a careful analysis is required, since the classical versions of Lyapunov's theorem [2], [6],

[14] cannot be used. They require a continuously differentiable, or at least a locally Lipschitz continuous Lyapunov function, but $V(x)$ (3) is continuous but not locally Lipschitz. However, as it is explained in the proof of Theorem 1 in Appendix A, it is possible to show the convergence properties by means of Zubov's theorem [17, Theorem 20.2, p. 568.], that requires only continuous Lyapunov functions.

Theorem 1: Consider the unperturbed system (1), with $\varrho_1 = \varrho_2 = 0$, and constant gains k_1, k_2 . Then the following statements are equivalent:

- i) The origin $x = 0$ of (1) is finite-time stable.
- ii) Matrix A (2) is Hurwitz, i.e., all its eigenvalues have negative real parts.
- iii) The constant gains are positive, i.e., $k_1 > 0, k_2 > 0$.
- iv) For every symmetric and positive definite matrix $Q = Q^T > 0$, the ALE (5) has a unique symmetric and positive definite solution $P = P^T > 0$.

In this case the function $V(x)$, defined in (3), is a global, *strict Lyapunov function* for system (1), for every $P = P^T > 0$ that is solution of the ALE (5) with an arbitrary matrix $Q = Q^T > 0$. Its time derivative $\dot{V}$ along the trajectories of the system satisfy the differential equation (4) almost everywhere. Moreover, a trajectory starting at x_0 at time $t = 0$ reaches the origin in a time smaller than $T(x_0)$, given by

$$T(x_0) = \frac{2}{\sigma} V^{1/2}(x_0), \quad \sigma = \frac{\lambda_{\min}\{P\} \lambda_{\min}\{Q\}}{\lambda_{\max}\{P\}} \quad (6)$$

where σ is a constant depending on the gains k_1, k_2 and Q .

III. LYAPUNOV FUNCTIONS FOR THE STA WITH PERTURBATIONS

It is well known [7], [9], [11], [12], that the STA (1) is robustly stable to perturbations globally bounded by

$$\varrho_1(x, t) = 0, \quad |\varrho_2(x, t)| \leq L \quad (7)$$

for any constant $L > 0$, when the gains k_1, k_2 are appropriately selected. In Theorem 2 this fact is established by means of a (robust) strict Lyapunov function, where robust means that it decreases monotonically along the trajectories of the system with perturbations.

Theorem 2: Suppose that the perturbation terms of system (1) are globally bounded by (7). Then for every positive value of $L > 0$ there exist gains k_1, k_2 such that the origin $x = 0$ is a robustly, globally finite-time stable equilibrium point. Moreover, there exists a symmetric and positive definite matrix $P = [p_{ij}]$ such that $V(x) = \zeta^T P \zeta$ is a quadratic, strict and robust Lyapunov function for the perturbed system, satisfying

$$\dot{V} \leq -|x_1|^{-1/2} \zeta^T Q_R \zeta \quad (8)$$

almost everywhere, for some symmetric and positive definite matrix $Q_R = [q_{Rij}]$. Furthermore, a trajectory starting at x_0 will converge to the origin in a finite time smaller than $\tilde{T}(x_0)$

$$\tilde{T}(x_0) = \frac{2}{\tilde{\sigma}} V^{1/2}(x_0), \quad \tilde{\sigma} = \frac{\lambda_{\min}\{P\} \lambda_{\min}\{Q_R\}}{\lambda_{\max}\{P\}}.$$

From the proof of Theorem 2 in Appendix B one can derive the following selection rules for the gains k_1, k_2 and for the matrices P and Q_R of the Lyapunov function.

Algorithm 1: i) Choose positive constants (β, γ) , so that $0 < \beta < 1, \gamma > 1$. ii) Find positive constants (κ, α) satisfying the inequality

$$\kappa - \frac{2}{\gamma} \alpha > \alpha^2 - \beta(1 + \kappa)\alpha + \frac{1}{4}(1 + \kappa)^2. \quad (9)$$

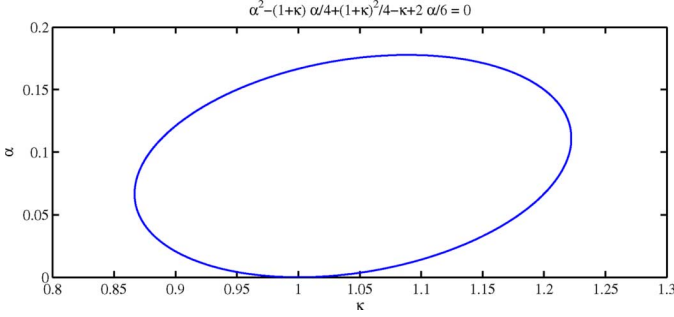


Fig. 1. Ellipse describing the boundary of the set (9) for $\beta = 1/4, \gamma = 6$.

This inequality represents the interior of an ellipse in the plane (κ, α) , parameterized by (β, γ) , with center at the point (see Fig. 1)

$$\alpha_c = \frac{\beta - \frac{1}{\gamma}}{1 - \beta^2}, \quad \kappa_c = \frac{(\beta - 2\frac{1}{\gamma})\beta + 1}{1 - \beta^2}. \quad (10)$$

If $\gamma\beta > 1$ it is always possible to find positive values of (κ, α) satisfying (9), since in this case the center (κ_c, α_c) of the ellipse is on the first quadrant, i.e., $\kappa_c > 0, \alpha_c > 0$. iii) Given such values of (β, γ) and (κ, α) the gains

$$k_1 = \kappa \sqrt{\frac{2\gamma}{(1-\beta)\alpha}} \sqrt{L}, \quad k_2 = \frac{(\beta+1)}{(1-\beta)} L \quad (11)$$

assure the robust, finite-time stability of the origin of the STA (1). iv) Matrices P and Q_R can be selected as

$$\begin{aligned} p_{11} &= 1, \quad p_{22} = \frac{(1-\beta)\alpha}{2L}, \quad p_{12} = -\sqrt{\frac{(1-\beta)\alpha}{2\gamma L}} = -\sqrt{\frac{p_{22}}{\gamma}}, \\ q_{R11} &= k_1 + 2p_{12}(k_2 + L) + 2L(1 - k_1 p_{12}) \frac{p_{22}}{p_{12}}, \\ q_{R12} &= -\frac{1}{2}(1 - k_1 p_{12}) + (k_2 + L)p_{22}, \quad q_{R22} = -p_{12}. \end{aligned} \quad (12)$$

Note that the set of gains for a given value of L obtained from Algorithm 1 is much bigger than the known recipes in [7], [9], [11], [12].

The next Theorem 3 shows that if the STA is robust against perturbations satisfying (7), then it will also endure perturbations with $\varrho_1(x, t) \neq 0$ that are globally bounded by

$$|\varrho_1| \leq \delta_1 + \delta_2 \sqrt{|x_1| + x_2^2}, \quad |\varrho_2| \leq L \quad (13)$$

for some non negative constants $\delta_1, \delta_2, L \geq 0$. If $\delta_1 = 0$, so that $\varrho_1(x, t)$ vanishes at the origin, and δ_2 is sufficiently small, then the trajectories will still converge to zero in finite time. If $\delta_1 \neq 0$, so that $\varrho_1(x, t)$ does not vanish at the origin, then the trajectories will (in general) not converge to the origin, but they will be *globally ultimately bounded* [8, Chapter 9], that is, there exists a positive constant b , and for every $a > 0$, there is $T = T(a, b) \geq 0$ such that

$$\|\zeta(t_0)\| \leq a \implies \|\zeta(t)\| \leq b, \quad \forall t \geq t_0 + T.$$

Theorem 3: Suppose that the perturbation terms of the system (1) are globally bounded by (13), and that the gains k_1, k_2 satisfy the conditions of the Theorem 2 for $\varrho_1(x, t) = 0, |\varrho_2(x, t)| \leq L$. Under these conditions the trajectories of system (1) are globally ultimately

bounded for δ_2 small enough, i.e., $\delta_2 < \lambda_{\min}\{Q_R\}/\eta$, and arbitrary $\delta_1 > 0$, with final bound

$$b = \sqrt{\frac{\lambda_{\max}\{P\}}{\lambda_{\min}\{P\}}} \frac{\delta_1 \eta}{(1-\kappa)(\lambda_{\min}\{Q_R\} - \delta_2 \eta)} \quad (14)$$

where $\eta \triangleq \sqrt{1 + p_{12}^2}$, $0 < \kappa < 1$, and P, Q_R are given in (12). Moreover, if $\delta_1 = 0$ then the origin $x = 0$ of (1) is an equilibrium point that is robustly globally finite-time stable, and all trajectories converge in a time smaller than $\tilde{T}(x_0)$

$$\tilde{T}(x_0) = \frac{2}{\tilde{\sigma}} V^{1/2}(x_0), \quad \tilde{\sigma} = \frac{\lambda_{\min}\{Q_R\} - \delta_2 \sqrt{1 + p_{12}^2}}{\lambda_{\max}^{1/2}\{P\}} \quad (15)$$

where x_0 is the initial state and $\tilde{\sigma}$ is a constant depending on the perturbation coefficient δ_2 .

IV. EXAMPLE

To illustrate the previous results consider a forced Van der Pol equation [8]

$$\dot{x}_1 = \varepsilon h(x_1) + x_2, \quad \dot{x}_2 = -x_1 + F(x, t)$$

with $h(x_1) = x_1 - x_1^3/3 + \phi(t)$, $\varepsilon > 0$, where $\phi(t)$ is an uncertain and bounded perturbation with $|\phi(t)| \leq \delta_1$ for all $t \geq 0$, and the forcing term $F(x, t)$ is also uncertain and bounded, i.e., $|F(x, t)| \leq L$ for all (x, t) . If $y = x_1$ is measured the following STA-based observer can be used to estimate x_2 :

$$\begin{aligned} \dot{\hat{x}}_1 &= \varepsilon h(y) + \hat{x}_2 - k_1 |\hat{x}_1 - y| \text{sign}(\hat{x}_1 - y), \\ \dot{\hat{x}}_2 &= -y - k_2 \text{sign}(\hat{x}_1 - y) \end{aligned}$$

The observation error, where $e_i = \hat{x}_i - x_i$ for $i = 1, 2$, is given by [see (1)]

$$\dot{e}_1 = -k_1 |e_1| \text{sign}(e_1) + e_2 - \phi(t), \quad \dot{e}_2 = -k_2 \text{sign}(e_1) - F(x, t).$$

In Fig. 2 simulation results are shown for the state, estimation and error trajectories. In Algorithm 1 selecting $\beta = 1/4, \gamma = 6$ and $L = 3$, the ellipse describing the boundary of the set (9) is represented in Fig. 1. Any point in its interior provides a suitable pair (κ, α) . Selecting the center point of the ellipse (10) $(\kappa_c, \alpha_c) = (1.044, 0.089)$ leads to the gains $k_1 = 24.27, k_2 = 5$. For the simulation $\varepsilon = 1$, $F(x, t) = 2.5 \sin(10\pi t)$ and $\phi(t) = 0$ for $t \in [0, 5]$ and switches to $\phi(t) = 2.5 \sin(20\pi t)$ for $t \in [5, 10]$. The states of the plant follow the typical limit cycle behavior, slightly perturbed by the presence of the perturbations $F(x, t)$ and $\phi(t)$. The Super-Twisting Observer estimation converges in finite-time to both states of the plant, despite of the unknown perturbation $F(x, t)$. However, when the perturbation term $\phi(t)$ appears at $t = 5$ s. the estimation errors are no longer zero but ultimately bounded.

V. CONCLUSION

In this technical note a method to construct a family of quadratic strict Lyapunov functions for the unperturbed and the perturbed STA has been provided, and for each element of the family an estimation of the reaching time is given. The freedom in the election of the Lyapunov function can be used for different purposes, as for example to optimize the reaching time estimation in the class of quadratic functions. Furthermore, the reaching time estimation relations are also useful to design the gains of the algorithm in such a way, that after a certain prefixed time the algorithm has converged, for a compact set of initial conditions.

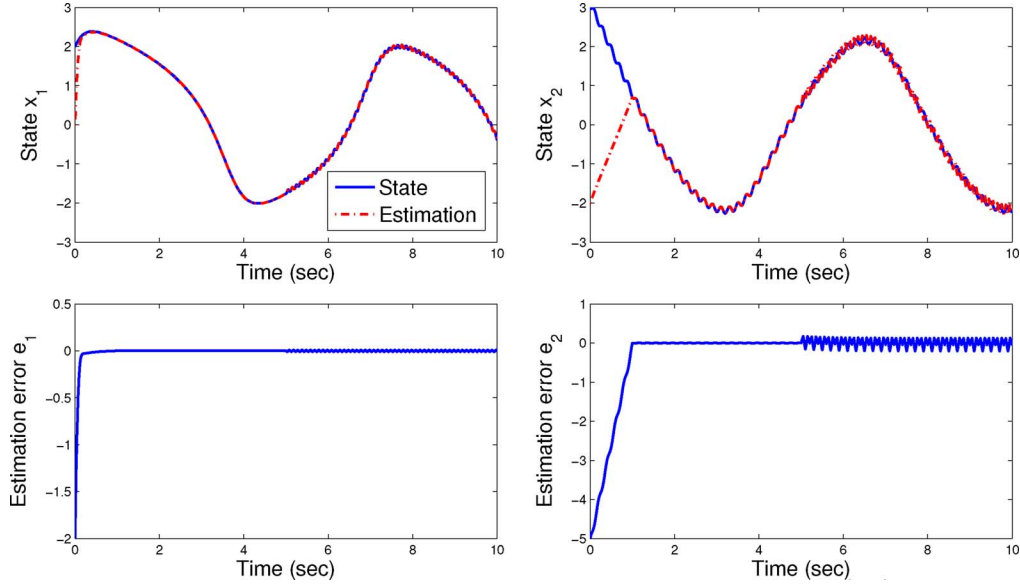


Fig. 2. Simulation of state, estimation and error trajectories for the Van der Pol system, for Plant's parameter $\varepsilon = 1$ and observer gains $k_1 = 24.27$, $k_2 = 5$. Unknown perturbations were $F(x, t) = 2.5 \sin(10\pi t)$ and $\phi(t) = 0$ for $t \in [0, 5]$ and switches to $\phi(t) = 2.5 \sin(20\pi t)$ for $t \in [5, 10]$.

APPENDIX

Proof of Theorem 1: Note first that $x = 0$ is an equilibrium point, since (1) is a differential inclusion $\dot{x} \in f(x)$, and $0 \in f(0)$ [2]. Equivalence of items (ii), (iii) and (iv) is a well-known fact from Lyapunov stability of Linear Time Invariant Systems [8]. Now it will be shown that (ii) $\implies$ (i). For any $Q = Q^T > 0$, with A Hurwitz, it follows that the ALE (5) has a unique solution $P = P^T > 0$. The proposed Lyapunov function $V(x) = \zeta^T P \zeta = p_{11}|x_1| + 2p_{12}x_2|x_1|^{1/2} \text{sign}(x_1) + p_{22}x_2^2$, where p_{ij} are the components of the matrix P , is an absolutely continuous (AC) function of x , and it is also positive definite and radially unbounded, i.e.

$$\lambda_{\min}\{P\}\|\zeta\|_2^2 \leq V(x) \leq \lambda_{\max}\{P\}\|\zeta\|_2^2 \quad (16)$$

where $\|\zeta\|_2^2 = |x_1| + x_2^2$ is the Euclidean norm of ζ . To ensure convergence to zero it is necessary to show that, along the trajectories $\varphi(t, x_0)$ of the system (1), $V(\varphi(t, x_0))$ decreases monotonically. When the Lyapunov function is continuously differentiable, or at least locally Lipschitz continuous, this is usually done by means of Lyapunov's theorem [2], [6], [8], [14]. However, $V(x)$ does not satisfy these conditions on the set $S = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 = 0\}$, due to the term $|x_1|^{1/2} \text{sign}(x_1)$. $V(x)$ can still be used as a Lyapunov function, in the same spirit as in the theorem of Zubov [17, Theorem 20.2, p. 568.], if we are able to show that it decreases monotonically along the trajectories of the system, and converges to zero.

If we are able to show that $V(\varphi(t, x_0))$ is AC as a function of time, then $V(\varphi(t, x_0))$ is a monotonically decreasing function if and only if $\dot{V}$ is negative definite *almost everywhere* [2, p. 207]. Note that $V(\varphi(t, x_0)) = V \circ \varphi(t, x_0)$ is the composition of $V(x)$, an AC function of x , and $\varphi(t, x_0)$, that by definition of a solution of the differential inclusion (1) is an AC function of time. However, the composition of two AC functions is in general *not* an AC function [3, p. 391]. Recall that the addition and product of two AC functions is always an AC function, but the composition of two (scalar) AC functions $h \circ g$ can be assured to be AC if either h is Lipschitz or g is monotone [3, p. 391]. In

$$\begin{aligned} V(\varphi(t, x_0)) &= p_{11}|\varphi_1(t, x_0)| \\ &+ 2p_{12}\varphi_2(t, x_0)|\varphi_1(t, x_0)|^{1/2} \text{sign}(\varphi_1(t, x_0)) + p_{22}\varphi_2^2(t, x_0) \end{aligned}$$

the problematic term to assure that $V(\varphi(t, x_0))$ is AC is the term $|x_1|^{1/2} \text{sign}(x_1) \circ \varphi_1(t, x_0) = |\varphi_1(t, x_0)|^{1/2} \text{sign}(\varphi_1(t, x_0))$, since $|x_1|^{1/2} \text{sign}(x_1)$ is not Lipschitz at $x_1 = 0$. However, as we will show in the next paragraph, $\varphi_1(t, x_0)$ is monotone when it goes through $\varphi_1(t, x_0) = 0$. This proves that $V(\varphi(t, x_0))$ is indeed an AC function of t , and therefore its derivative is defined almost everywhere and it is given by (4), where it is defined.

In order to show that $\varphi_1(t, x_0)$ is monotone when crossing zero, suppose that $\varphi_2(\tau, x_0) \neq 0$ at the instant $t = \tau$, when $\varphi_1(t, x_0)$ crosses zero. Then from the differential inclusion $\dot{x}_1 \in -k_1|x_1|^{1/2} \text{sign}(x_1) + x_2$ it follows that $\varphi_1(t, x_0)$ will be monotonically increasing or decreasing during an interval containing τ . If $\varphi_2(t, x_0) = 0$ when $\varphi_1(t, x_0)$ crosses zero, then $\varphi_1(t, x_0)$ will stay in zero. In both cases $V(\varphi(t, x_0))$ is AC.

Since $V(\varphi(t, x_0))$ is AC and $\dot{V}(4)$ is n.d. almost everywhere, it follows that $V(\varphi(t, x_0))$ is a monotonically decreasing function [2, p. 207]. Moreover, from (16) and the fact that $|x_1|^{1/2} \leq \|\zeta\|_2 \leq \lambda_{\min}^{-1/2}\{P\}V^{1/2}(x)$ it follows that $\dot{V}(\varphi(t, x_0)) \leq -\sigma V^{1/2}(\varphi(t, x_0))$, where σ is given by (6). Since an AC function is the integral of its derivative [3], then

$$\begin{aligned} V(\varphi(t, x_0)) - V(\varphi(0, x_0)) &= \int_0^t \dot{V}(\varphi(\tau, x_0)) d\tau \\ &\leq -\sigma \int_0^t V^{1/2}(\varphi(\tau, x_0)) d\tau. \end{aligned}$$

Bihari's inequality [17, p. 509] implies that $V(\varphi(t, x_0)) \leq (V^{1/2}(x_0) - (\sigma/2)t)^2$, so that $V(\varphi(t, x_0))$ and $\varphi(t, x_0)$ converge to zero in a finite time smaller than $T(x_0)$ given in (6).

We show that (i) $\implies$ (ii), or equivalently, that $\sim$ (ii) $\implies \sim$ (i). Assume that A is not Hurwitz, so we will show that $x = 0$ is not asymptotically stable for (1). Several cases are possible here, depending on the position of the eigenvalues $\lambda_{1,2} = -k_1/4 \pm \sqrt{k_1^2/16 - k_2/2}$ of matrix A .

- 1) $\lambda_{1,2}$ are imaginary, i.e., $k_1 = 0, k_2 > 0, P = \text{diag}\{k_2, 1\} > 0$, and $Q = 0$ are solutions of (5), so that $V(x)$ is locally Lipschitz, p.d. and $\dot{V} = 0$. This means that $V(\varphi(t, x_0))$ remains constant along the solutions of the system, and so it does not converge to zero.

- 2) $\lambda_{1,2} = 0$, i.e., $k_1 = 0$, $k_2 = 0$. A has two eigenvalues at the origin. In this case the system is linear, and the origin is clearly unstable.
- 3) $\lambda_{1,2}$ are real with different signs, i.e., $k_1^2 \geq 8k_2$ and $k_2 < 0$. Then

$$P_c = \begin{bmatrix} -k_1 & 1 \\ 1 & 0 \end{bmatrix}, Q_c = \begin{bmatrix} 2k_2 - k_1^2 & k_1 \\ k_1 & -1 \end{bmatrix} < 0$$

are solutions of (5). An extension of Chetaev's Instability Theorem [8, Theorem 4.3] will be used to prove instability. For this consider Chetaev's function $V_c(x) = \zeta^T P_c \zeta$, that is positive in the set $U = \{\zeta \in \mathbb{R}^2 \mid \zeta_1(2\zeta_2 - k_1\zeta_1) > 0\}$. U contains points arbitrarily near to the origin, and has a boundary at $\{\zeta_1 = 0\}$ and $\{2\zeta_2 = k_1\zeta_1\}$, where $V_c(x) = 0$. For any initial point $x_0 \in U \cap B_r$, with $B_r = \{\zeta \in \mathbb{R}^2 \mid \|\zeta\|_2 \leq r\}$, $V_c(x_0) > 0$, and, since $\dot{V}_c(x) = \zeta^T Q_c \zeta > 0$ in U , the trajectory $\varphi(t, x_0)$ has to leave $U \cap B_r$ through the boundary of B_r . This implies that the origin is unstable.

- 4) One of the eigenvalues $\lambda_{1,2}$ is zero, i.e., $k_1 \neq 0$ and $k_2 = 0$. It follows that $\zeta_2(t) = \zeta_2(0)$ for all $t \geq 0$, so that no asymptotic stability is possible.
- 5) $\lambda_{1,2}$ have both positive real part, so that matrix A is Anti-Hurwitz. In this case the ALE (5) has a positive definite solution P for every negative definite matrix Q , and therefore $V(x) > 0$, $\dot{V}(x) > 0$. Instability follows from the previous extension of Chetaev's theorem.

Proof of Theorem 2: Consider a quadratic Lyapunov function candidate $V(x) = \zeta^T P \zeta$ (3), with P given in (12). If the perturbation satisfy (7) then $\varphi_1(t, x_0)$ is monotone when it crosses zero, and so $V(\varphi(t, x_0))$ is AC and its time derivative along the solutions of system (1), when it exists, is given by

$$\dot{V} = -|x_1|^{-1/2} \zeta^T Q \zeta + \frac{2}{|x_1|^{1/2}} \left[\frac{1}{2} \varrho_1, |x_1|^{1/2} \varrho_2 \right] P \zeta. \quad (17)$$

Using (7) and setting $\nu(t, x) = \varrho_2(t, x) \text{sign}(x_1)$, (17) becomes

$$\dot{V} = |x_1|^{-1/2} \zeta^T \left[(A + N(t, x))^T P + P(A + N(t, x)) \right] \zeta, \\ \text{where } A + N(t, x) = \begin{bmatrix} -\frac{1}{2}k_1 & \frac{1}{2} \\ -(k_2 - \nu(t, x)) & 0 \end{bmatrix}.$$

Since $\nu(t, x)$ is bounded $V(x)$ is p.d. and $\dot{V}$ is n.d. if

$$p_{22} > p_{12}^2, p_{12} < 0, 0 > k_1 p_{12} + \frac{1}{4}(1 - k_1 p_{12})^2 + \dots \\ \dots + 2p_{12}^2(k_2 - \nu(t, x)) - (1 - k_1 p_{12})p_{22}(k_2 - \nu(t, x)) + \dots \\ \dots + (k_2 - \nu(t, x))^2 p_{22}^2. \quad (18)$$

Using the fact that $(k_2 - L) \leq (k_2 - \nu(t, x)) \leq (k_2 + L)$, and noticing that $k_2 > L$ has to be satisfied (otherwise the origin would be unstable for the perturbation $\nu(t, x) = L$), the last inequality in (18) will be satisfied if

$$0 > k_1 p_{12} + \frac{1}{4}(1 - k_1 p_{12})^2 + 2p_{12}^2(k_2 + L) - \\ - (1 - k_1 p_{12})p_{22}(k_2 - L) + (k_2 + L)^2 p_{22}^2$$

or, equivalently, that the matrix Q_R given in (12) is p.d., i.e., $Q_R > 0$. It follows that (18) will be satisfied if the inequalities

$$\gamma > 1, \kappa > 0, 0 < \beta < 1, \alpha > 0, \\ \kappa - \frac{2}{\gamma}\alpha > \alpha^2 - \beta(1 + \kappa)\alpha + \frac{1}{4}(1 + \kappa)^2 \quad (19)$$

are satisfied, where

$$\kappa \triangleq -k_1 p_{12}, \gamma \triangleq \frac{p_{22}}{p_{12}^2}, \alpha \triangleq p_{22}(k_2 + L), \beta \triangleq \frac{(k_2 - L)}{(k_2 + L)}. \quad (20)$$

As discussed in the Algorithm 1 the last inequality in (19) represents the interior of an ellipse in the plane (κ, α) , parameterized by (β, γ) . If $\gamma\beta > 1$ it is always possible to select a point (κ, α) such that all inequalities (19) are satisfied. In particular, this will happen for the center point (κ_c, α_c) (10) of the ellipse. Given a quadruple $(\beta, \gamma, \kappa, \alpha)$ that satisfies (19), the set of gains (11) and the values of the components of P (12) are directly derived from (20). The rest of the proof is analogous to the proof of Theorem 1.

Proof of Theorem 3: From Theorem 2 it is known that $V(x) = \zeta^T P \zeta$ is a Lyapunov function for the (structured) perturbed system (with $\varrho_1(x, t) = 0$, $|\varrho_2(x, t)| \leq L$), and (8) is satisfied for P and Q_R in (12). $\dot{V}$ along the solutions of the perturbed system (1), when it exists, is given by

$$\dot{V} \leq -\frac{1}{|x_1|^{1/2}} \left\{ \zeta^T Q_R \zeta - [\varrho_1 \quad 0] P \zeta \right\} \\ \leq -\frac{1}{|x_1|^{1/2}} \left\{ \lambda_{\min} \{Q_R\} \|\zeta\|_2^2 - (\delta_1 + \delta_2 \|\zeta\|_2) \eta \|\zeta\|_2 \right\} \\ \leq -\frac{\|\zeta\|_2}{|x_1|^{1/2}} \left\{ (\lambda_{\min} \{Q_R\} - \delta_2 \eta) \|\zeta\|_2 - \delta_1 \eta \right\} \\ \leq -(\lambda_{\min} \{Q_R\} - \delta_2 \eta) \|\zeta\|_2 + \delta_1 \eta$$

where $\eta \triangleq \sqrt{1 + p_{12}^2}$ and the fact that $|x_1|^{1/2} \leq \|\zeta\|_2$ has been used. When δ_2 is small, i.e.

$$\delta_2 < \frac{\lambda_{\min} \{Q_R\}}{\eta} \quad (21)$$

it follows that for every $\delta_1 > 0$ and $0 < \kappa < 1$:

$$\dot{V} \leq -(\lambda_{\min} \{Q_R\} - \delta_2 \eta) [\kappa \|\zeta\|_2 + (1 - \kappa) \|\zeta\|_2] + \delta_1 \eta \\ \leq -\frac{\kappa(\lambda_{\min} \{Q_R\} - \delta_2 \eta)}{\lambda_{\max}^{1/2} \{P\}} V^{1/2}(x), \forall \|\zeta\|_2 \geq \mu, \\ \text{where} \\ \mu \triangleq \frac{\delta_1 \eta}{(1 - \kappa)(\lambda_{\min} \{Q_R\} - \delta_2 \eta)}.$$

Therefore any trajectory enters the set $\Omega_\mu = \{x \in \mathbb{R}^2 \mid V(x) \leq \lambda_{\max} \{P\} \mu^2\}$ in finite time T , that using Bihari's inequality (proof of Theorem 1) can be calculated for an initial state x_0 as

$$T = \frac{2\lambda_{\max}^{1/2} \{P\}}{\kappa(\lambda_{\min} \{Q_R\} - \delta_2 \eta)} \left[V^{1/2}(x_0) - \lambda_{\max}^{1/2} \{P\} \mu \right]$$

and stays there for all future times, so that $\|\zeta(t)\|_2 \leq b$ for all $t \geq t_0 + T$, with b as in (14). If $\delta_1 = 0$ and δ_2 satisfies (21), as in the proof of Theorem 1, finite-time convergence, with convergence time estimated by (15), follows.

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Multiple Model Adaptive Mixing Control: The Discrete-Time Case

Simone Baldi, Petros Ioannou, and Edoardo Mosca

Abstract—Most of the work in multiple model adaptive control with various forms of switching focused on continuous-time systems. The purpose of this technical note is to extend the results of one approach, the adaptive mixing control (AMC), to discrete-time systems. Further, the technical note solves the tracking problem which has not been addressed in most schemes of this class. Stability and robustness properties of the AMC scheme for discrete-time systems are analyzed. It is shown that in the ideal case, when no disturbances or unmodeled dynamics are present, the tracking error converges to zero. In the non ideal case, the mean-square tracking error is of the order of magnitude of the modeling error provided the unmodeled dynamics satisfy a norm-bound condition. While these robustness results are conceptually similar to those of traditional robust adaptive control, the proposed scheme does not suffer from the drawback of stabilizability of the estimated plant and in addition performs much better in simulation studies. Furthermore, it allows well developed results from robust control to be incorporated in the design.

Index Terms—Multiple model adaptive control, robust adaptive control.

I. INTRODUCTION

All real systems are affected by uncertainties due to unmodeled dynamics, unknown parameters and process changes. A practical control design must be able to maintain performance and stability irrespective of these uncertainties. Adaptive control was motivated to meet the challenge of handling parametric uncertainties much larger than those robust control can handle, when the uncertainties are so large to the point that no single fixed LTI controller can guarantee satisfactory closed-loop behavior. This is typically achieved by estimating the unknown parameters on-line and using the estimates as the true ones in calculating the controller parameters. For the on-line calculation of the controller parameters to be possible at all times, the estimated plant has to satisfy the assumptions of stabilizability and detectability at each instant of time. Such a property is not guaranteed by the traditional adaptive control schemes. In order to enforce it, a number of different modifications and new classes of adaptive controllers have been proposed [1]–[3]. A class of adaptive controllers which bypasses the stabilizability/detectability problem and can take advantage of the robust control tools for LTI systems involves a supervisory logic based adaptive control architecture which consists of two main components: (1) a *multicontroller* formed by a family of candidate controllers designed off-line to meet the control objectives based on different possible sets of values of the plant parameters; and (2) a *supervisor* responsible of deciding which controller or combination of controllers

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S. Baldi was with the Dipartimento Sistemi e Informatica, Università di Firenze, Firenze 50139, Italy, and is now with the Computer Science Department, University of Cyprus (UCY), Nicosia 1678, Cyprus (e-mail: sbaldi@dsi.unifi.it).

P. Ioannou is with the Department of Electrical Engineering, University of Southern California, Los Angeles, CA 90089 USA (e-mail: ioannou@usc.edu).

E. Mosca is with Dipartimento Sistemi e Informatica, Università di Firenze, Firenze 50139, Italy (e-mail: mosca@dsi.unifi.it).

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